DS Practical - 5

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Binary Search Tree: Write a program to:

a. Create a binary search tree.

CODE :

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
rootA = TreeNode('A')
rootB = TreeNode('B')
rootC = TreeNode('C')
rootD = TreeNode('D')
rootE = TreeNode('E')
rootF = TreeNode('F')
rootG = TreeNode('G')

root.left = rootA
root.right = rootB

rootA.left = rootC
rootA.right = rootD

rootB.left = rootE
rootB.right = rootF

rootF.left = rootG

print("root.right.left.data:", root.right.left.data)
print("\nKOMAL S049")
```

OUTPUT :

```
root.right.left.data: E

KOMAL S049
```

b. Insert nodes into a binary search tree.

CODE :

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
rootA = TreeNode('A')
rootB = TreeNode('B')
rootC = TreeNode('C')
rootD = TreeNode('D')
rootE = TreeNode('E')
rootF = TreeNode('F')
rootG = TreeNode('G')
rootH = TreeNode('H')

root.left = rootA
root.right = rootB

rootA.left = rootC
rootA.right = rootD

rootB.left = rootE
rootB.right = rootF

rootF.left = rootG
rootF.left = rootH

print("root.right.left.data:", root.right.left.data)

print("\nKOMAL S049")
```

OUTPUT :

```
root.right.left.data: E

KOMAL S049
```

c. Search for a node in a binary search tree.

CODE :

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
rootA = TreeNode('A')
rootB = TreeNode('B')
rootC = TreeNode('C')
rootD = TreeNode('D')
rootE = TreeNode('E')
rootF = TreeNode('F')
rootG = TreeNode('G')
rootH = TreeNode('H')

root.left = rootA
root.right = rootB

rootA.left = rootC
rootA.right = rootD

rootB.left = rootE
rootB.right = rootF

rootF.left = rootG
rootF.right = rootH

def search(root, key):
    if root is None:
        return False

    print("Checking:", root.data)

    if root.data == key:
        return True

    return search(root.left, key) or search(root.right, key)

key = input("Enter element to search: ")

if search(root, key):
    print("Element Found")
else:
    print("Element Not Found")

print("\nKOMAL S049")
```

OUTPUT :

```
================ RESTART: C:/
Enter element to search: A
Checking: R
Checking: A
Element Found

KOMAL S049
```

```
Enter element to search: K
Checking: R
Checking: A
Checking: C
Checking: D
Checking: B
Checking: E
Checking: F
Checking: G
Checking: H
Element Not Found

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```